PITWALL

Algorithms and Mathematics of a 1v1 F1
Opponent-Prediction and Counter-Strategy Engine

November 2026

Abstract

PITWALL estimates tyre degradation from public lap times, predicts when an opponent will pit, and simulates our options to the end of the race to recommend one. It chains five simple, explainable models rather than one large one. This document sets out the mathematics of each model, the assumptions behind it, and how the system was evaluated, including where it falls short. A live terminal runs the same models during a race and is scored against real results from unseen 2025 and 2026 races. Meta's Muse Spark then answers an engineer's questions about each call, using only the system's numbers. It explains the call and cannot change it.

1 System overview

The pipeline runs once per lap and may use only laps that have already been completed.

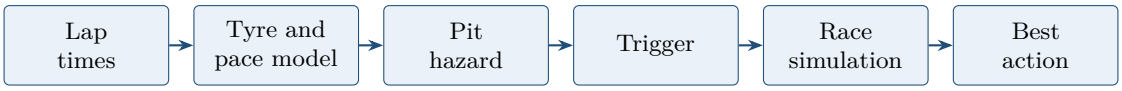


Figure 1: The decision pipeline. Here the simulation runs only when the trigger fires; the live terminal in Section 8 always runs it.

Notation. t is the current lap, T the race distance, g the gap in seconds (positive when we trail), and L_{pit} the time lost to a pit stop.

2 Tyre and pace model

2.1 Model

Each clean lap time is decomposed into a stint baseline, a tyre penalty, a fuel effect and noise:

$$L_t = B_s + D(\text{age}_t) + F_t + \varepsilon_t, \qquad F_t = 0.04\,(T - t), \tag{1}$$

where the tyre penalty is a non-negative, non-decreasing curve:

$$D(\text{age}) = b_1 a + b_2 a^2, \qquad a = \frac{\text{age} \times c_{\text{compound}}}{10}, \qquad b_1, b_2 \geq 0. \tag{2}$$

The assumed compound factors are $c_{\text{SOFT}} = 1.4$, $c_{\text{MEDIUM}} = 1.0$ and $c_{\text{HARD}} = 0.7$.

2.2 Why this form

- **Fuel is assumed, not fitted.** In a first stint, tyre age and fuel load change in perfect lockstep (correlation -1), so lap times alone cannot separate them.
- **One baseline per stint.** Otherwise a compound change would be read as tyre wear.
- **Non-negative coefficients.** Tyres cannot get faster with age.

2.3 Estimation

Lap 1, in-laps, out-laps, neutralised laps and outliers more than 3 s from the stint median are removed, and at least 6 clean laps are required. With $y_t = L_t - F_t$, the coefficients minimise a ridge objective, with $\lambda = 0.1$ on the curve terms:

$$\min_{b_1, b_2 \geq 0,\, B} \sum_t \left(y_t - B_{s(t)} - b_1 a_t - b_2 a_t^2\right)^2 + \lambda\,(b_1^2 + b_2^2). \tag{3}$$

The constraint has only four possible active sets (both terms, b_1 only, b_2 only, neither), so each is solved in closed form and the best valid fit is kept.

2.4 Forecast uncertainty

Forecast variance combines lap noise with uncertainty in the coefficients:

$$\mathrm{Var}[\hat{L}] = \sigma^2 + x^\top C x, \qquad C = \sigma^2 \left(X^\top X + \Lambda\right)^{-1}, \tag{4}$$

where x holds the regressors for the forecast lap, X the regressors for all fitted laps, and Λ the ridge penalties.

3 Pit-stop prediction: discrete-time hazard model

3.1 Hazard

The hazard is the probability of stopping on lap t, given that the car has reached that lap, and is modelled by logistic regression:

$$h_t = P(\text{stop at end of lap } t \mid \text{reached lap } t,\, X_t) = \frac{1}{1 + e^{-\beta^\top X_t}}. \tag{5}$$

3.2 Features

There are 16 inputs, and all of them use only laps before t. This restriction matters: an earlier version counted the stop being predicted among its inputs, which produced falsely excellent scores.

Group	Features
Tyres	age, age^2, age×soft, age×hard, soft, hard
Race progress	lap fraction, lap fraction^2, final 5 laps indicator
Stops	no stop yet indicator, stops completed
Degradation	estimated penalty D, estimated slope D'
Race control	safety car or VSC or red flag, its interaction with age
Constant	bias

3.3 Training

The coefficients minimise the L_2-penalised logistic loss by Newton's method:

$$\beta \leftarrow \beta - \left(Z^\top W Z + \Lambda\right)^{-1} \left(Z^\top (p - y) + \Lambda \beta\right), \qquad W = \mathrm{diag}\big(p_i(1 - p_i)\big), \tag{6}$$

where Z is the standardised design matrix and Λ applies a penalty of 1 to all terms except the bias. Stops are not reweighted, so outputs remain probabilities.

3.4 From hazards to a forecast

Projecting tyre age forward gives hazards for the next five laps, which combine into:

$$P(\text{stop within } k \text{ laps}) = 1 - \prod_{j=1}^{k} (1 - h_{t+j}), \tag{7}$$

$$P(\text{first stop on lap } t+j) = h_{t+j} \prod_{i=1}^{j-1} (1 - h_{t+i}). \tag{8}$$

The remaining probability is assigned to "no stop within the window".

Example. Hazards of 14%, 17% and 20% give $P(\text{within } 3) = 1 - 0.86 \times 0.83 \times 0.80 \approx 43\%$.

4 Trigger

The simulation runs if any of these conditions holds:

$$\text{opponent just stopped} \quad \text{or} \quad P(\text{stop within } 3) \geq \theta \quad \text{or} \quad \text{loss} \geq 3 \text{ s}, \tag{9}$$

where the threat fades to zero once the cars are a full pit stop apart:

$$\text{loss} = P(\text{stop within } 3) \times D_{\text{own}} \times 3 \times \max\Big(0,\ 1 - \frac{|g|}{L_{\text{pit}}}\Big). \tag{10}$$

5 Race simulation

5.1 Pit loss

Pit loss is learned only from stops already seen in the race, using the median across both cars of

$$c = L_{\text{in}} + L_{\text{out}} - 2\,\tilde{L}, \tag{11}$$

where $\tilde{L}$ is the car's median clean lap so far. Until a stop has been observed, a prior of 22 s is used. Real timing data places about 15% of the loss on the in-lap and 85% on the out-lap, so a car that has just boxed still owes the rest:

$$g_{\text{eff}} = g + 0.85\,L_{\text{pit}}\big(\mathbf{1}_{\text{we boxed}} - \mathbf{1}_{\text{they boxed}}\big). \tag{12}$$

5.2 Candidate actions

The candidates are *pit now*, *pit in 1, 2 or 3 laps*, and *stay out*. Invalid actions are removed: no stop on the final lap, no stop straight after a stop, and no staying out if a required stop could then no longer be made.

5.3 Planning later stops

After the first decision, each car follows its cheapest remaining plan, found by dynamic programming over lap and tyre age:

$$V(h, a) = D(a) + \min\Big\{\, V(h+1,\, a+1),\ \ L_{\text{pit}} + V(h+1,\, 1) \,\Big\}. \tag{13}$$

At the end of the race $V = 0$, or $V = \infty$ if the dry-race rule requiring a second compound has not been met.

5.4 Monte Carlo rollout

Simulated lap times for each car are

$$\hat{L}_h = B + D(\text{age}_h) + F_h + \eta_h + \delta\,\mathbf{1}_{\text{after stop}} + \epsilon\,\text{age}_h + \text{pit cost}_h, \tag{14}$$

with independent draws $\eta_h \sim \mathcal{N}(0, \sigma^2)$ for lap noise, $\delta \sim \mathcal{N}(0, 0.4^2)$ for the unknown next compound, and $\epsilon \sim \mathcal{N}\big(0, \min(\text{SE}_{\text{slope}}, 0.03)^2\big)$ for uncertainty in the tyre curve.

The gap then evolves lap by lap, subject to an overtaking rule. A car that catches the other is held 0.6 s behind unless it is at least 1.0 s faster on that lap. Stops are exempt, so cars can swap places in the pits. This rule is what makes undercuts and overcuts possible, because the opponent's timing now determines who rejoins ahead.

Algorithm 1 Evaluating the candidate actions from lap t

```
Draw shared random numbers for N = 1000 races, including each opponent stop
for each valid action A do
    Build our stop plan: the first stop from A, the rest by dynamic programming
    for each simulated race do
        Build the opponent's plan from its sampled stop
        g ← g_eff
        for each remaining lap h do
            g' ← g + L̂_h^own − L̂_h^opp
            if no pit lap and the chaser catches the other car and is under 1.0 s faster then
                Hold the chasing car 0.6 s behind
            end if
            g ← g'
        end for
        Record the final gap G = −g
    end for
    Compute P(G > 0), E[G] and the 5th percentile of G
end for
```

Every action reuses the same random draws (*common random numbers*), so differences between actions reflect the decision and not sampling luck.

6 Decision

Each action A receives a utility, with seconds scaled by the pit loss so that all terms are comparable:

$$U(A) = P(G > 0) - 0.25\,\min\Big(1, \frac{\max(0, -\mathbb{E}[G])}{L_{\text{pit}}}\Big) - 0.10\,\min\Big(1, \frac{\max(0, -G_{5\%})}{L_{\text{pit}}}\Big). \tag{15}$$

The highest utility is recommended. Differences under 1% count as ties and are settled by the larger expected gap, so a car already clear does not take a pointless stop.

7 Own car versus opponent

Both cars use the same pace model, so neither gets a built-in advantage. The difference is information: in the synthetic race, our tyre curve is fitted to a private wear signal that stands in for team telemetry. This signal is synthetic and is never presented as measured data. Historical races have no such signal, so both cars use public lap times.

8 The live terminal

The terminal (`python -m pitwall.live`) runs the same models once per line crossing and at once on any flag change. Unlike the pipeline above, it always simulates, so our own plan stays current when the rival is no threat. The trigger only raises alerts. The values below were learned from 2023–24 races, and the changes apply only to the terminal.

8.1 Stop cost

The prior is the median green-flag stop at that circuit, and ranges from about 19 s to 29 s. Once three stops have been seen in this race, the median across the whole field replaces it. A safety car or VSC makes a stop cheaper, but it may end before the car boxes, so the expected cost of boxing at the end of the lap h laps from now is

$$c_h = L_{\text{pit}}\Big(1 - (1 - r_k)\, q_k^{h+1}\Big), \qquad q_k = \frac{\sum_i \max(d_i - 1,\, 0)}{\sum_i d_i}, \tag{16}$$

where r_k is the measured cost share, and q_k is the chance that a period of kind k still runs one lap later, from past period lengths d_i in laps. The measured values are $r_{\text{SC}} = 0.673$, $q_{\text{SC}} = 0.737$, $r_{\text{VSC}} = 0.812$ and $q_{\text{VSC}} = 0.377$.

8.2 Tyre wear

Early in a stint, lap times often show no wear. The fit is therefore pulled towards the curve b^0 this circuit showed in past races, with $\lambda = 10$:

$$\min_{b \ge 0,\, B} \sum_t \big(y_t - B_{s(t)} - b_1 a_t - b_2 a_t^2\big)^2 + \lambda \,\|b - b^0\|^2. \tag{17}$$

Drivers also manage pace, so a tyre can look healthy until it is finished. Past the 90th percentile stint length A_c for that compound and circuit, an assumed cliff applies:

$$D_{\text{live}}(\text{age}) = D(\text{age}) + 0.25\, \max(0,\ \text{age} - A_c). \tag{18}$$

The hazard model keeps the plain fit it was trained on.

8.3 Pace gap

A new stint's baseline rests on a few noisy laps, and over many laps its error swamps the gap. With Δ the gap in the latest stint baselines, m the gap over the race so far (weighted by laps), and n the clean laps in the latest stint, the simulated gap is

$$\Delta' = m + \kappa\,(\Delta - m), \qquad \kappa = \frac{\tau^2}{\tau^2 + \sigma_{\text{own}}^2/n_{\text{own}} + \sigma_{\text{opp}}^2/n_{\text{opp}}}. \tag{19}$$

Fitting on the training races drove τ to the bottom of its range, so in practice $\Delta' \approx m$: the race-long pace gap predicted real results better than the latest stint.

8.4 Displayed win chance

Raw simulated probabilities were too confident, so the screen shows a blend with the gap signal z, fitted by logistic regression on the training races:

$$\text{logit}(P_{\text{win}}) = 0.72\ \text{logit}\big(P(G > 0)\big) + 0.234\, z, \qquad z = \frac{-g_{\text{eff}}}{0.75\sqrt{T - t}}, \tag{20}$$

with z clipped to ± 6. All options in a decision share the same z and the first coefficient is positive, so the blend cannot change their order by $P(G > 0)$. Ranking still uses the utility U from the raw simulation.

8.5 Rejoin position

If we box at the end of lap b, our time at the end of the out-lap is compared with every running car i, which is assumed to hold its recent pace $\tilde{L}_i$ without stopping:

$$R_{\text{own}} = R^{\text{line}}_{\text{own}} + (b + 1 - t)\,\tilde{L}_{\text{own}} + c, \qquad R_i = R^{\text{line}}_i + (b + 1 - t_i)\,\tilde{L}_i, \tag{21}$$

where R^{line} is the clock time of a car's last line crossing, on lap t_i, and c is the stop cost. The predicted position is one plus the number of cars with $R_i < R_{\text{own}}$. A car less than 2 s ahead on the road, lapped or not, is reported as traffic.

8.6 Keeping calls stable

A call that changes with sampling noise is unusable on the radio, and "box in 1 lap" followed by "box now" is one call, so calls are tracked by the lap to box on.

- **A stop must earn the call.** Waiting keeps later options open, so a stop is chosen over staying out only if it gains at least 0.02 utility or 1 s at the flag.
- **A held call must be beaten.** It changes only for a gain of at least 0.03 utility, or when a car stops or the flag changes.
- **A rival behind does not delay the call.** After 3 s their lap is projected from their pace, then corrected when they cross the line.

9 Explaining calls with Muse Spark

The statistical engine makes every forecast and call. Muse Spark 1.3, reached through the Meta Model API, answers the engineer's questions about them. Information flows one way, from a finished decision to the model, so no answer can alter a call.

9.1 Keeping answers on the facts

- **Facts only.** Each question is sent with the current race state as JSON. The model must reply `NOT IN DATA` to anything the facts do not answer.
- **Sums supplied.** The positions after a stop are computed by the engine. If we stop now, with no stop from the rival, the gap becomes $g + L_{\text{now}}$. If the rival stops and we do not, it becomes $g - L_{\text{now}}$. In testing, given $g = 3.1$ s and $L_{\text{now}} = 22.9$ s, the model answered 19.8 s behind instead of 26.0 s.
- **Numbers checked.** Let F be the numbers in the facts and the question. A number n quoted to d decimal places passes if

$$\min_{f \in F} |n - f| \le 0.5 \times 10^{-d}, \tag{22}$$

so rounding 22.9 to 23 passes, while 22.5 does not. An answer that fails gets one retry and is then withheld.

- **Never in the way.** Requests run on a background thread, so a slow reply or a failure leaves the calls unaffected. Without a key, the terminal works the same.

The model runs at `minimal` reasoning: `low` and `medium` used up a 600-token limit on reasoning and returned no answer.

10 Evaluation

10.1 Pit-stop prediction on unseen races

The model was trained on 41 dry races from 2023–24 and tested on 22 dry races from 2025 (23,764 car-laps and 685 stops), so no test race influenced training. The scores are

$$\text{Brier} = \frac{1}{n}\sum_i (p_i - y_i)^2, \qquad \text{Log loss} = -\frac{1}{n}\sum_i \left[y_i \ln p_i + (1 - y_i)\ln(1 - p_i)\right], \tag{23}$$

where lower is better for both.

Per-lap predictor	Brier	Log loss
PITWALL hazard model	**0.0274**	**0.1199**
Tyre-age-only baseline	0.0278	0.1274
Constant base rate	0.0280	0.1308

Table 1: Scores on 22 unseen races from 2025.

At $\theta = 30\%$, the three-lap forecast warned before 34% of 636 stops and fired on 6.7% of laps with no stop coming. Probabilities are reasonably calibrated up to about 30% and run high above that: predictions of 40–50% came true 25% of the time. The advantage over the baselines is real but small.

10.2 Strategy in simulated races

Each policy drove the same 300 randomised races with the same noise.

Policy	P(ahead at flag)	Mean gap (s)	Mean stops
PITWALL	**70%**	**+17.2**	**2.24**
PITWALL ignoring the opponent forecast	70%	+16.9	2.32
Tyre-age rule	50%	+4.4	2.00
Reactive (cover their stop)	44%	+2.6	1.85
Fixed one-stop	35%	−9.8	1.00

Table 2: Paired comparison over 300 simulated races.

PITWALL's average of 2.24 stops is close to the true optimum of about 2.2, which was checked on 40 of these races. Two caveats apply:

- **The opponent forecast adds no measurable benefit here** (0 points, 95% interval ±1.7). The advantage comes from modelling our own tyres and planning to the flag. The simulated opponent never reacts to us, which limits what predicting it can add.

- **The test favours PITWALL.** These races share the simulator's overtaking rule, and PITWALL receives the synthetic wear signal. The results show good decisions within the model's world, not proof that it would win real races.

10.3 Live terminal on unseen real races

Each race was played through the terminal's engine and compared with what really happened: 21 dry races from 2025 and 18 from 2026, with four pairs of cars per race chosen by position at lap 5. Settings were fitted on 2023–24 only. Nothing was fitted on the test races, but looking at their scores led to the win-chance blend and the race-long pace gap. Win chance is scored for the option the team really took, since the others never happened.

Measure	2025	2026
Win chance, Brier score	**0.111**	**0.129**
Gap alone, $\Phi(z)$ (normal CDF)	0.129	0.170
"Current order holds"	0.155	0.219
Coin flip	0.250	0.250
Next-lap pace, mean error	0.47 s	0.48 s
Pace 10 laps ahead on the same tyres, mean error	0.69 s	0.67 s
Rejoin position within 2 places (mean error)	76% (1.6)	75% (1.8)
Rival stops flagged in the 3 laps before	35%	43%
Call changes per pair of cars	4.7	5.0

Table 3: Live terminal scored against real results. Lower Brier is better.

Screen says	0–10%	10–30%	30–50%	50–70%	70–90%	90–100%
Came true, 2025	3%	16%	38%	71%	86%	98%
Came true, 2026	4%	29%	46%	67%	81%	95%

Table 4: How often the displayed win chance came true.

A decision takes about 5 ms at 1000 simulations per option (worst 58 ms over 353 decisions), against a lap of about 80 s. In the larger scoring run at 300 simulations, the slowest decisions took 2.7 s (2025) and 1.3 s (2026), and the cause has not been investigated. Four caveats apply:

- **The gap alone wins in the last 10 laps**: 0.042 against 0.057 in 2025, and 0.026 against 0.039 in 2026.
- **Absolute finish gaps remain poor.** The mean error is about 14 s, no better than assuming the gap stays as it is, so the screen shows only time between options.
- **Most rival stops come without warning.** More than half are not flagged.
- **Calls cannot be scored directly.** Teams made other choices, so nothing here shows that a call would have gained a place. Five laps before a real stop, the planned stop lap is a median 4 laps away from it.

10.4 Muse Spark answers on unseen races

Facts came from nine decisions in three 2026 races, with six questions each through the real API: four with a known numeric answer, one asking why, and one the facts cannot answer.

Measure	Result
Known-value questions answered correctly	36 of 36
Unanswerable questions refused	9 of 9
Explanations returned	9 of 9
First attempts quoting a number not in the facts	0 of 54
Answers withheld, API errors	0, 0
Response time, median (worst)	2.2 s (6.7 s)

Table 5: Muse Spark 1.3 at minimal reasoning, 54 questions.

Three caveats apply:

- **The sample is small,** and the known-value test only requires the right number to appear in the reply.
- **The number check never had to step in during this run,** so its retry and withholding are covered only by tests with fake replies. It also cannot judge reasoning: a wrong statement built from correct numbers would pass.
- **Explanations mostly restate the engine's numbers.** They are short and accurate, but they compare win chance and time rather than add strategic reasoning.

11 Assumptions and limitations

Assumed	Learned from data
Fuel effect (0.04 s/lap)	Tyre curve and stint baselines, per car
Compound wear factors (1.4, 1.0, 0.7)	Pit-stop coefficients β
Overtaking rule (1.0 s advantage, 0.6 s following)	Pit loss, once a stop has been seen
Utility weights (0.25, 0.10)	Lap noise σ
New-tyre pace uncertainty ($\pm$0.4 s)	
In-lap share of pit loss (15%, measured on 12 real stops)	Live: circuit stop costs, wear curves, stint norms
Live: tyre cliff (0.25 s per lap past the norm)	Live: safety car and VSC costs, win-chance blend

Table 6: What the system assumes and what it learns.

Known limitations:

- The opponent does not react to our stop, and the hazard model does not use the gap to us.
- Future safety cars, and the way one bunches the field up, are not simulated. The race simulation is still two cars: the live terminal reports traffic and rejoin position beside it. Outside the terminal, a stop under a safety car costs a full pit loss.
- The terminal checks which compounds normally reach the flag, but does not know which sets a team has left or race one compound against another.
- The live OpenF1 connection has run on real rows from finished races and on a local test server, but not yet during a running session.
- The models cover dry running only. In the wet, the engine stays in monitoring mode and says why.

- In the dashboard and demo, absolute gaps run wide, because current pace is held to the flag. In the 2023 Bahrain race, Alonso versus Verstappen forecasts about $-100\,\mathrm{s}$ at lap 13, while the real final gap was about $-38\,\mathrm{s}$. Options should be compared with each other rather than read as predicted finishing gaps.
- Alternative outcomes are simulated counterfactuals, never observed history.